WHAT IS TURBULENCE?

Key features, examples, and why it matters

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IBFE Tutorials • Module 5

Adapted from a lecture by Steven L. Brunton.

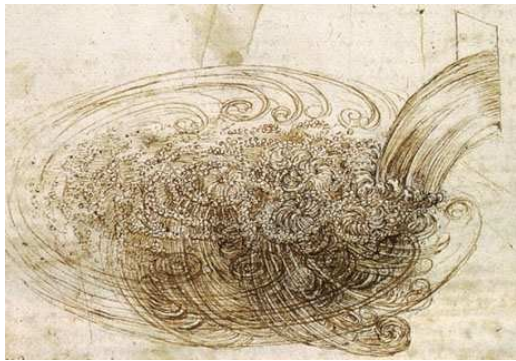
A working viewpoint

Turbulence is recognized through a collection of dynamical signatures, not one universal checklist.

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

WHY TURBULENCE FASCINATES US

- Turbulence shows up in **art** and **language**: roiling, churning, turbulent.
- We have strong intuition for “turbulent” vs “not,” but a single universal definition is hard.
- So: we focus on **key features** and **canonical examples**.



Leonardo da Vinci, *Studies of water*, c. 1510–1512 (Royal Collection, public domain).

TURBULENCE IN ART: VAN GOGH



Vincent van Gogh, *The Starry Night* (1889), MoMA (public domain).

A PRACTICAL “FINGERPRINT” OF TURBULENCE

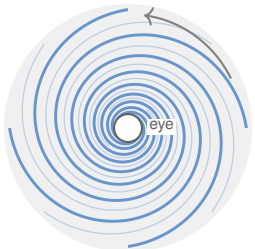
A useful operational fingerprint is that turbulent flows are typically:

1. **Three-dimensional** (3D vortical motion),
2. **Multi-scale** in space and time (big eddies $\rightarrow$ small eddies),
3. **Unsteady** (time-evolving),
4. **Mixing** and strongly transporting scalar/particles.

Mental experiment: add dye upstream $\Rightarrow$ rapid stretching and folding $\Rightarrow$ enhanced mixing. This is a diagnostic, not a formal definition.

A NOTE ON 2D VS 3D TURBULENCE

- In 3D turbulence, kinetic energy is typically transferred toward smaller scales and ultimately dissipated by viscosity.
- Idealized 2D turbulence has two inviscid quadratic invariants: energy tends upscale, while enstrophy tends downscale.
- Rotating or stratified geophysical flows can be **quasi-2D**; finite thickness, friction, and wave dynamics modify the ideal picture.

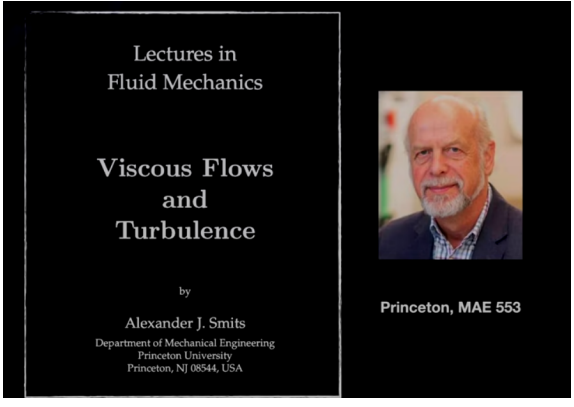


cyclone: $L \sim 500$ km

Northern-Hemisphere tropical cyclone seen from above: counterclockwise rotation, with spiral rain bands that trail the rotation (inflow spirals inward cyclonically). Its horizontal scale ($\sim$300–1000 km) far exceeds the $\sim$10–16 km depth of the troposphere, which is why such flows are quasi-2D. Schematic: instructor's course notes.

A RECOMMENDED REFERENCE (FROM THE LECTURE)

- Lex Smits (Princeton): **MAE 553 Viscous Flows and Turbulence** notes.
- Use as a companion source for canonical flows and experimental intuition.



Title page of A. J. (Lex) Smits, *MAE 553: Viscous Flows and Turbulence* lecture notes (Princeton University).

EXAMPLE: A TURBULENT BOUNDARY LAYER (SIMULATION)

- High-fidelity simulation of a developing boundary layer (think: flow over a wing/car).
- Observe: **3D**, **unsteady**, **multi-scale**, **mixing**.

3D Simulation (YouTube)

HALLMARK: MULTI-SCALE STRUCTURE

- “Zoom in anywhere” $\Rightarrow$ you see more structure.
- Multi-scale content is a quick visual diagnostic of turbulence.

Fun aside (from lecture): blockbuster explosions look “high budget” when they show many scales.



Still from *Oppenheimer* (dir. C. Nolan, Universal Pictures, 2023); shown for educational commentary, rights reserved by the studio.

TURBULENCE AT PLANETARY SCALES

- Ocean currents and atmospheric patterns are highly turbulent.
- Turbulent transport $\Rightarrow$ global mixing of heat, salt, nutrients, pollutants.



Photo: Kent Smith, Gulf of Alaska (2007), glacial meltwater meeting ocean water.

RAYLEIGH–BÉNARD CONVECTION AND CHAOS

- Heated from below, cooled from above $\Rightarrow$ convective instability.
- Unsteady + multi-scale + mixing (often used to motivate chaotic dynamics).
- Historical link: Lorenz (1963) studied simplified models of convective rolls.

3D Simulation (YouTube)

MATHEMATICAL ENTRY POINT: INCOMPRESSIBLE NAVIER–STOKES

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}}\,\Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

$$\mathbf{u}(\mathbf{x}, t) = \begin{bmatrix} u(\mathbf{x}, t) \\ v(\mathbf{x}, t) \\ w(\mathbf{x}, t) \end{bmatrix}$$

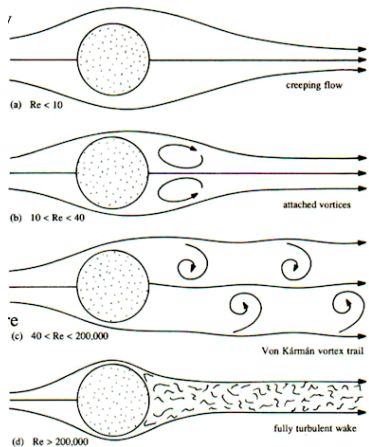
- $\boldsymbol{u}(\boldsymbol{x}, t) \in \mathbb{R}^3$ is a 3-component velocity field.
- Rich behavior emerges as parameters/forcing/geometry change.

REYNOLDS NUMBER: A CONTROL KNOB FOR COMPLEXITY

- For chosen characteristic scales U and L, the **Reynolds number** compares inertial and viscous effects:

$$\mathrm{Re} \sim \frac{UL}{\nu}.$$

- It is a **control parameter**, but geometry, forcing, roughness, and disturbances also determine the observed state.
- In high-Re, 3D, locally isotropic turbulence, Kolmogorov scaling estimates $L/\eta \sim \mathrm{Re}_L^{3/4}$; this is not a universal one-to-one law.

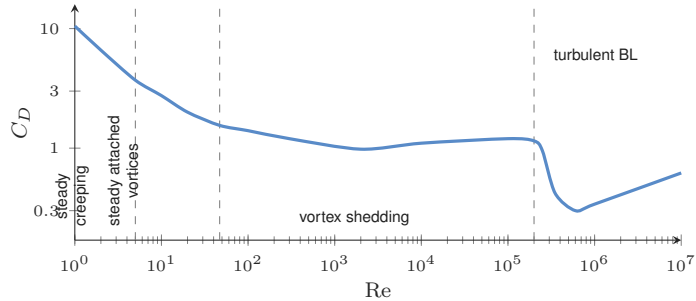


Flow regimes past a cylinder; after S. Vogel, *Life in Moving Fluids*, 2nd ed. (Princeton Univ. Press, 1994) (figure number to confirm).

The panel boundaries (10, 40, 200,000) are rounded; the next slide marks the more precise values $\mathrm{Re} \approx 5,\ 47$, and 2×10^5.

DRAG DEPENDS STRONGLY ON REYNOLDS NUMBER

- As flow regime changes with Re, the drag coefficient can change dramatically.
- Implication: “just changing speed” can change force laws in big ways.



Drag coefficient of a circular cylinder versus Reynolds number with flow regimes. Schematic (instructor’s course notes).

Dashed lines: $\mathrm{Re} \approx 5$ (steady separation, a recirculating eddy pair forms), $\mathrm{Re} \approx 47$ (onset of periodic shedding, a Hopf bifurcation; the wake itself becomes turbulent for $\mathrm{Re} \gtrsim 200$–300), $\mathrm{Re} \approx 2 \times 10^5$ (onset of the drag crisis: the boundary layer turns turbulent before separating, so C_D falls from ≈ 1.2 to ≈ 0.3 near 3–5×10^5). Curve is qualitative; values after standard compilations (e.g. Schlichting & Gersten; Williamson 1996).

ASIDE: GALILEO'S BALL DROP THROUGH MULTIPLE REGIMES

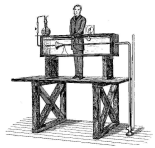
- Falling ball accelerates $\Rightarrow$ effective Re increases during the fall.
- Drag force can be highly unsteady and regime-dependent.
- Heavy objects: turbulent drag may be negligible relative to weight; light objects: not.
- The gravitational field g may be nearly constant, while the ball's acceleration need not remain g once drag is appreciable.



Portrait of Galileo Galilei (public domain); the legendary experiment at Pisa.

WHY “REYNOLDS”? PIPE-FLOW EXPERIMENTS (1883)

- Osborne Reynolds observed changes in pipe flow as speed and disturbance conditions varied:
  - laminar (layered, steady),
  - transitional,
  - turbulent (roiling, disordered).
- The laminar solution exists at every Re; pipe transition is subcritical and depends on finite-amplitude disturbances.
- Pipe flow remains a core engineering problem: pumping water, sewage, and process fluids efficiently.
- Textbooks quote $\mathrm{Re}_D \approx 2300$ for the onset of transition in pipes; Reynolds’ own critical values are on the next slide.



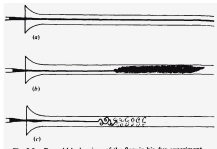


Fig. 9.2. Reynolds's drawings of the flow in his dye experiment.

Apparatus and dye-filament plate from O. Reynolds, *Phil. Trans. R. Soc. Lond.* **174** (1883), 935–982 (public domain).

PIPE FLOW: WHICH "CRITICAL" REYNOLDS NUMBER?

Here $\mathrm{Re}_D = U_b D/\nu$ uses the bulk (mean) velocity U_b and the pipe diameter D.

- Textbooks quote $\mathrm{Re}_D \approx 2300$ for the onset of transition in pipes.
- Reynolds (1883) kept the flow laminar up to $\mathrm{Re} \approx 13{,}000$ (his "upper critical" value) only with a very still reservoir and a smooth bell-mouth inlet.
- With disturbances present, turbulence did not persist below $\mathrm{Re} \approx 2000$ (his "lower critical" value).
- Later, quieter experiments stayed laminar to $\mathrm{Re} \sim 10^5$ (Pfenninger 1961): there is no upper limit set by a linear instability.
- Deck 5-4 discusses the modern statistical value $\mathrm{Re}_D \approx 2040$ (Avila et al. 2011), where puff splitting balances puff decay.

> The lower value is a property of the flow (below it, turbulence decays whatever the disturbance); the upper value is a property of the experiment (how quiet the inlet is).

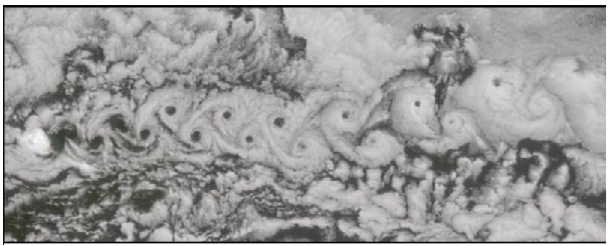
TRANSITION TO TURBULENCE

- A classical experiment is to inject dye into a tube to visualize the flow.
- As Re and/or disturbance amplitude increase, the dye filament may become unsteady and then turbulent.
- A single critical value is not universal: inlet noise, roughness, and protocol matter.

Pipe flow transition to turbulence (YouTube)

TURBULENCE IS NOT “RANDOM”: COHERENT STRUCTURES

- Turbulence has stochastic aspects, but it is not structureless noise.
- **Coherent structures** (organized patterns) can persist within complex flows.
- Similar large-scale patterns can appear across wildly different Reynolds numbers.

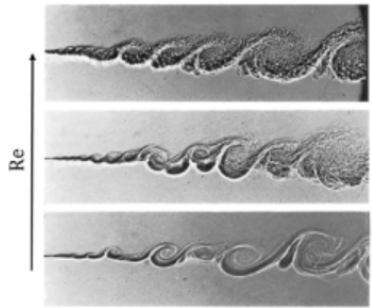




M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982).

MIXING LAYERS: LAMINAR VS TURBULENT SHARE PATTERNS

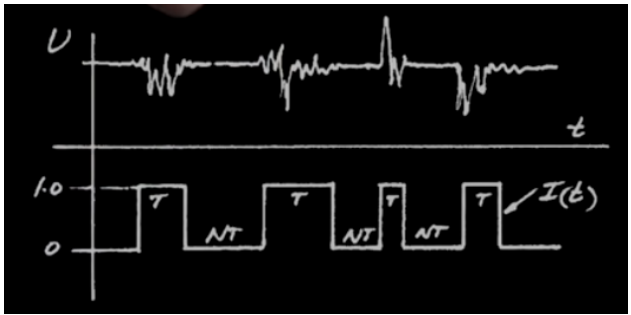
- Classic mixing-layer images show:
  - Bottom: low-Re (laminar),
  - Middle: intermediate-Re (transitional), and
  - Top: high-Re (turbulent).
- All cases share recognizable, coherent structures.
- Turbulence adds scales and distortion, but does not erase organization.



Shadowgraphs of a plane mixing layer at increasing Reynolds number; G. L. Brown & A. Roshko, *J. Fluid Mech.* **64**, 775–816 (1974); reproduced in Van Dyke, *An Album of Fluid Motion*.

ANOTHER HALLMARK: INTERMITTENCY

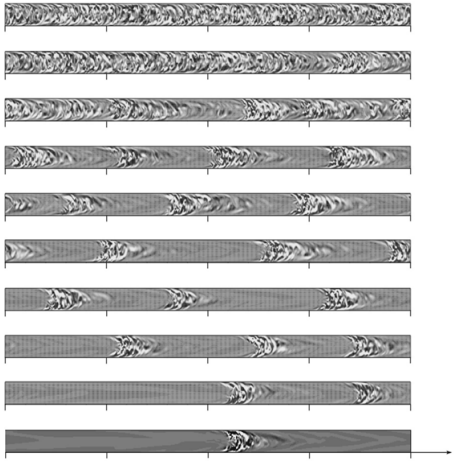
- Turbulence can be **patchy** in space/time:
  - turbulent region $\rightarrow$ calm region $\rightarrow$ turbulent again.
- Picture: smoke plume, clouds in the boundary layer, bursts of turbulence.
- Useful concept: an "intermittency function" that switches on/off turbulence.



Source: Smith and Smits, 1988.

Image is a still of a lightboard sketch from S. L. Brunton's lecture (title slide); the signal $u(t)$ and intermittency function $I(t)$ ($I = 1$ in turbulent "T" intervals, 0 in non-turbulent "NT" ones) follow the cited Smith & Smits example (full reference to be confirmed). The time average of I is the intermittency factor γ.

INTERMITTENCY IN PIPE FLOW

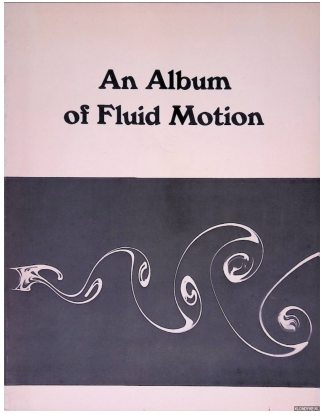


Source: Nikitin, N. & Pimanov, V. (2015). Numerical study of localized turbulent structures in a pipe. *Fluid Dynamics* 50, 655–664.

Panels range from uniformly turbulent pipe flow (top) to localized turbulent patches (puffs) separated by laminar flow (bottom): intermittency in space as well as in time. See deck 5-4 for puff decay and splitting.

HOW WE HANDLE COMPLEXITY: CANONICAL FLOWS

- Canonical flows: isotropic turbulence, wall-bounded flows, boundary layers, mixing layers, pipe flow, channel flow, cavity flow, wakes, jets.
- Strategy: distill turbulence into idealized building blocks.
- Real systems combine many of these (boundary layer + wake + jet + mixing layer).

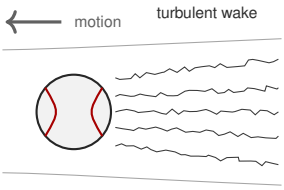


M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982), cover.

EXAMPLES OF CANONICAL FLOWS AND WHY THEY MATTER

Wakes

- drag and unsteady forces
- vehicles, trains, projectiles, fish, birds

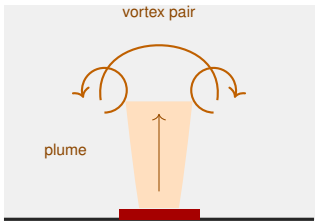


$\mathrm{Re} \sim 2 \times 10^5$

Schematic (instructor's course notes).
Baseball wake: $D \approx 7.4$ cm at 40 m/s in air gives $\mathrm{Re} = UD/\nu \approx 2 \times 10^5$, near the drag crisis.

Mixing layers

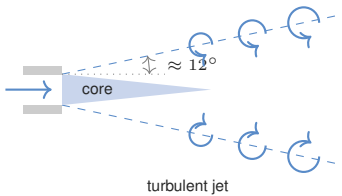
- shear-driven mixing
- ocean waves, atmospheric inversion layers



Schematic (instructor's course notes).
Starting thermal plume: its edges are shear (mixing) layers that entrain ambient fluid; the cap is a vortex ring, seen in section as a counter-rotating pair.

Jets

- thrust and power
- noise and quality-of-life impacts



Schematic (instructor's course notes).
$\approx 12^\circ$ is the visible outer edge; the half-velocity radius grows more slowly, $\mathrm{d}r_{1/2}/\mathrm{d}x \approx 0.09$–$0.10$ (≈ 5–6°; Pope 2000, §5.1). The potential core ends ≈ 4–5 diameters downstream.

END GOAL: PREDICT, OPTIMIZE, AND CONTROL

- Engineering and scientific aim: develop models that predict how outcomes change with design knobs.
- Examples: increase lift, decrease drag, enhance mixing, reduce noise.
- Control: actively manipulate flow for better performance.



Source: Ball, P. Wing and fin motions share universal principles. Nature (2014).

PREVIEW: REYNOLDS DECOMPOSITION

Recall the Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}} \Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

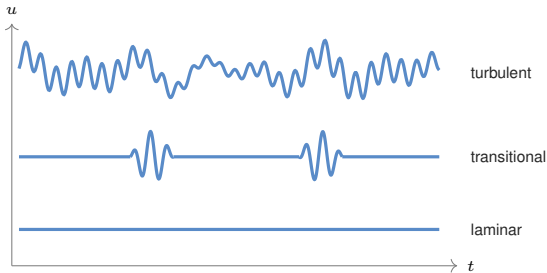
Choose an averaging operator and write the Reynolds decomposition:

$$\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}, t) + \boldsymbol{u}'(\boldsymbol{x}, t), \qquad \overline{\boldsymbol{u}'} = \boldsymbol{0}.$$

- The mean may be an ensemble, spatial, phase, or time average.
- For statistically stationary, ergodic flow, a sufficiently long time average can represent the ensemble mean.
- Fluctuations $\boldsymbol{u}'$: turbulent deviations around the mean.

WHY AVERAGING BECAME NATURAL HISTORICALLY

- Before modern field measurements (e.g., PIV/laser sheets), data were often pointwise.
- Hot-wire anemometry: long time series at one point $\rightarrow$ build statistics.
- Averaging is both practical (data) and useful (reduced modeling complexity).



Sketch of laminar, transitional, and turbulent velocity signals recorded at a point (e.g. by a hot-wire anemometer). Schematic (instructor's course notes).

TAKEAWAYS

- Turbulence is hard to define, but easy to recognize by its **fingerprint**:

$$\text{3D} \;+\; \text{multi-scale} \;+\; \text{unsteady} \;+\; \text{mixing}.$$

- The Navier–Stokes equations remain the anchor; Re is an important control parameter, not a complete state label.
- Coherent structures and intermittency mean turbulence is **not** pure randomness.
- Canonical flows are how we build understanding and models.
- What's next: Reynolds averaging/closure.

SELECTED REFERENCES

- O. Reynolds, “An experimental investigation of the circumstances which determine whether the motion of water shall be direct or sinuous,” *Phil. Trans. R. Soc. Lond.* **174** (1883), 935–982. doi:10.1098/rstl.1883.0029.
- A. N. Kolmogorov, “The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers,” *Proc. R. Soc. A* **434** (1991 translation), 9–13. doi:10.1098/rspa.1991.0075.
- R. H. Kraichnan, “Inertial ranges in two-dimensional turbulence,” *Phys. Fluids* **10** (1967), 1417–1423. doi:10.1063/1.1762301.
- K. Avila, D. Moxey, A. de Lozar, M. Avila, D. Barkley, and B. Hof, “The onset of turbulence in pipe flow,” *Science* **333** (2011), 192–196. doi:10.1126/science.1203223.
- S. B. Pope, *Turbulent Flows*, Cambridge University Press (2000).

SELECTED REFERENCES (CONTINUED)

- S. L. Brunton, lecture on turbulence (source of the slide sequence of this deck), doi:10.52843/cassyni.hrpvdf.
- A. J. Smits, *MAE 553: Viscous Flows and Turbulence*, lecture notes, Princeton University, PDF.
- M. Van Dyke, *An Album of Fluid Motion*, Parabolic Press (1982).
- G. L. Brown and A. Roshko, "On density effects and large structure in turbulent mixing layers," *J. Fluid Mech.* **64** (1974), 775–816. doi:10.1017/S002211207400190X.
- S. Vogel, *Life in Moving Fluids*, 2nd ed., Princeton University Press (1994).
- C. H. K. Williamson, "Vortex dynamics in the cylinder wake," *Annu. Rev. Fluid Mech.* **28** (1996), 477–539. doi:10.1146/annurev.fl.28.010196.002401.
- H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed., Springer (2017) (cylinder drag curve and drag crisis).
- E. N. Lorenz, "Deterministic nonperiodic flow," *J. Atmos. Sci.* **20** (1963), 130–141. doi:10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2.
- W. Pfenninger, "Boundary layer suction experiments with laminar flow at high Reynolds numbers in the inlet length of a tube by various suction methods," in G. V. Lachmann (ed.), *Boundary Layer and Flow Control*, Pergamon (1961), 961–980.
- N. V. Nikitin and V. O. Pimanov, "Numerical study of localized turbulent structures in a pipe," *Fluid Dynamics* **50** (2015), 655–664.
- Image credits for the paintings, film still, photographs, and the Ball (2014, *Nature*) and Smith & Smits (1988) figures are given on the slides where they appear.